

# PROFESSIONAL AGRICULTURAL HANDBOOK

## Crop Diseases of Pakistan

Comprehensive Management Guide for 20+ Diseases Across 12 Major Crops

2025 Edition - 100+ Pages

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A comprehensive resource for farmers, extension officers, and agricultural professionals on identifying, diagnosing, and managing major crop diseases affecting Pakistan's agriculture. Optimized for AI integration and knowledge base embedding.

*Includes: Disease biology | Pathogen taxonomy | Symptomology | Epidemiology | Control strategies | Resistant varieties  
| FAQs | Seasonal calendars*

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## 1. INTRODUCTION & PAKISTAN AGRICULTURAL CONTEXT

### Agriculture in Pakistan

Pakistan's agriculture sector contributes approximately 20% to national GDP and employs over 38% of the population. The country is the 8th largest wheat producer globally and a major producer of cotton, rice, sugar cane, vegetables, and fruits. However, crop diseases remain a significant impediment to achieving food security and sustaining rural livelihoods. Annual yield losses due to diseases range from 10-25%, representing millions of metric tons of lost production and billions in economic losses.

### Climate Zones and Agro-Ecology

Pakistan spans diverse agro-ecological zones: Punjab (moderate to semi-arid, 5-45°C), Sindh (semi-arid to arid, hot summers 40-45°C), Khyber Pakhtunkhwa (cool highlands, 5-30°C), and Balochistan (arid). These climatic variations significantly influence disease prevalence and management strategies. Temperature fluctuations, monsoon rainfall patterns (June-September), and seasonal variations define risk periods for different diseases.

### Economic Impact of Crop Diseases

Unmanaged crop diseases cause significant economic losses: wheat rusts alone cost 500 million USD annually; potato late blight causes 10-15 billion USD globally; cotton leaf curl virus caused 20 billion USD losses in Pakistan (1992-1997). Beyond production loss, disease-affected produce has reduced market value, limited export opportunities, and shortened shelf-life. Farmers incur costs for fungicides, labor, and lost income.

### How to Use This Handbook

This comprehensive handbook provides detailed information on 20+ major crop diseases affecting Pakistan's agriculture. Each disease chapter covers: pathogen biology and taxonomy, disease cycle and transmission, symptomology at different severity levels, epidemiology and risk factors, diagnosis methods, and integrated management strategies. Chapters are structured for optimal readability and AI-based knowledge base integration.

# WHEAT — STRIPE RUST (YELLOW RUST)

Disease Code: WR001 | Severity: VERY HIGH

## 1. DISEASE OVERVIEW

Stripe rust (also known as yellow rust) is one of the most destructive diseases of wheat in Pakistan. It appears as narrow, elongated pustules arranged in stripes on leaf surfaces.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Punjab, Sindh, KP

## 2. HISTORY & PAKISTAN CONTEXT

First documented in Pakistan in 1920s. Has caused major epidemics affecting millions of hectares. Major outbreak in 1974-75 caused 30% yield loss nationally.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                             |
|-----------------|-------------------------------------|
| Scientific Name | Puccinia striiformis f. sp. tritici |
| Disease Code    | WR001                               |
| Common Names    | N/A                                 |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne uredospores; long-distance wind-dispersed; can overwinter on wheat residue

**Optimal Conditions:** Temperature: 8-15°C (optimal 12°C); High humidity; Cool, moist conditions

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                        |
|----------------|---------------------------------------------------------------------------------|
| Early Stage    | Small chlorotic lesions on leaf blades, initially yellow-green                  |
| Moderate Stage | Elongated, parallel yellow pustules on upper leaf surface; leaves may yellow    |
| Severe Stage   | Dense striping on leaves and stems; complete defoliation; premature plant death |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties: AARI-2011, Shahkar-2013, Borlaug-2016
- Monitor fields from December onwards
- Apply fungicides: Propiconazole (25% EC) at 800ml/ha
- Crop rotation with non-cereal crops
- Destroy infected crop residue

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** How early can rust appear?

**A:** Stripe rust can appear as early as December in wheat crop

**Q:** What is economic threshold?

**A:** 5% flag leaf area infected or >10% leaf area infected on lower leaves

**Q:** Can I mix fungicides?

**A:** Yes, but avoid mixing with alkaline products. Follow label instructions

**Q:** When is critical control period?

**A:** From boot stage to milk stage - early fungicide application is crucial

**Q:** Are organic controls available?

**A:** Limited; sulfur dust can provide partial control, resistance breeding is most reliable

# WHEAT — LEAF RUST (BROWN RUST)

Disease Code: WR002 | Severity: HIGH

## 1. DISEASE OVERVIEW

Leaf rust appears as circular to oval pustules on leaf surfaces, usually brown in color. More common than stripe rust in Pakistan.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All regions

## 2. HISTORY & PAKISTAN CONTEXT

Historically more prevalent than stripe rust. Major epidemics recorded in 1990s and 2000s.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                    |
|-----------------|----------------------------|
| Scientific Name | Puccinia triticina Erikss. |
| Disease Code    | WR002                      |
| Common Names    | N/A                        |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Uredospores airborne; local spread through rain splash; alternate hosts may be involved

**Optimal Conditions:** Temperature: 15-21°C (optimal); Requires moisture on leaves for infection

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                     |
|----------------|------------------------------------------------------------------------------|
| Early Stage    | Small circular lesions with halo; reddish-brown coloring                     |
| Moderate Stage | Circular to oval rust pustules on leaf surfaces; leaves may become chlorotic |
| Severe Stage   | Heavy pustule development; extensive necrosis; leaf senescence               |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Deploy resistant varieties: PK-2013, Anaj-2017, Tameer-2019
- Early monitoring (January-February)
- Fungicide application: Azoxystrobin (25% SC) at 600ml/ha
- Nitrogen management - excessive nitrogen increases susceptibility
- Field sanitation of crop residue

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** What's the difference with stripe rust?

**A:** Leaf rust has round pustules; stripe rust has linear arrangement

**Q:** Is leaf rust less serious?

**A:** Equally serious economically; can cause 20-40% yield loss

**Q:** When to start monitoring?

**A:** January onwards at weekly intervals

**Q:** Can both rusts occur together?

**A:** Yes, mixed infections complicate management

**Q:** What rain conditions favor infection?

**A:** Light rain with high humidity; heavy rain may wash off spores

# RICE — BLAST DISEASE (LEAF BLAST, NECK BLAST)

Disease Code: RB001 | Severity: VERY HIGH

## 1. DISEASE OVERVIEW

Rice blast is the most destructive fungal disease globally. Causes diamond-shaped lesions on leaves and devastating damage on panicles.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All rice regions, Particularly Punjab and Sindh

## 2. HISTORY & PAKISTAN CONTEXT

Major disease in Pakistan since 1960s. Causes recurring epidemics. Major outbreak in 2004 affected 400,000 hectares.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                        |
|-----------------|------------------------------------------------|
| Scientific Name | Pyricularia oryzae Cavara (Magnaporthe oryzae) |
| Disease Code    | RB001                                          |
| Common Names    | N/A                                            |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; rain-splash water dispersal; contaminated seed; infected crop debris

**Optimal Conditions:** Temperature: 22-28°C (optimal); High humidity (95-100%); Free moisture on leaves (8+ hours)

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                                         |
|----------------|--------------------------------------------------------------------------------------------------|
| Early Stage    | Small gray-green lesions with narrow red-brown borders on leaves                                 |
| Moderate Stage | Lesions expand to diamond-shaped spots; center becomes gray; borders remain reddish-brown        |
| Severe Stage   | Multiple lesions coalesce; entire leaf blades wither; panicles rot at neck causing complete loss |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use blast-resistant varieties: HR-12, PR-114, BH-160
- Proper seedling management - avoid excessive crowding in nursery
- Apply fungicides: Tricyclazole (75% WP) at 600g/ha or Validamycin A (3% L) at 2-3 L/ha
- Timing: First spray at panicle initiation, second at boot stage
- Proper drainage to reduce humidity
- Balanced fertilization - avoid excessive nitrogen

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** What is neck blast?

**A:** Infection at panicle base causes panicle death - most damaging form

**Q:** When is peak risk period?

**A:** Panicle initiation to heading stage (maximum vulnerability)

**Q:** How much yield loss possible?

**A:** 30-100% loss possible; complete crop failure in severe cases

**Q:** Can resistant varieties be overcrowded?

**A:** No; still need proper spacing to reduce humidity

**Q:** Is early variety management needed?

**A:** Yes; healthy nursery establishment is crucial for disease prevention

# RICE — BACTERIAL LEAF BLIGHT

Disease Code: RB002 | Severity: HIGH

## 1. DISEASE OVERVIEW

Bacterial leaf blight causes elongated lesions with yellow halos on rice leaves. Major disease in tropical and subtropical regions.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Punjab, Sindh

## 2. HISTORY & PAKISTAN CONTEXT

Emerged as significant problem in Pakistan in 1990s. Now present in all major rice growing areas.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                       |
|-----------------|-------------------------------|
| Scientific Name | Xanthomonas oryzae pv. oryzae |
| Disease Code    | RB002                         |
| Common Names    | N/A                           |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Bacteria in plant debris; insect vectors; water splash; contaminated tools

**Optimal Conditions:** Temperature: 24-32°C; High humidity; Free moisture on leaves; Plant wounds facilitate entry

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                             |
|----------------|----------------------------------------------------------------------|
| Early Stage    | Waterlogged-appearing lesions with yellow halo starting at leaf tips |
| Moderate Stage | Lesions elongate along veins; may coalesce; leaves turn yellow       |
| Severe Stage   | Extensive leaf blight; stem infection; premature plant death         |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties: TJ-3055, PK-6129, Basmati-385
- Avoid overhead irrigation - use furrow irrigation
- Remove diseased plant parts immediately
- Apply copper-based bactericides: Copper Oxychloride (50% WP) at 2.5 kg/ha
- Sanitize harvesting tools
- Seed treatment: Seed soak in hot water (50°C for 30 min)

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Is this curable once present?

**A:** No; management focuses on prevention and containment

**Q:** Do resistant varieties guarantee protection?

**A:** Resistance can break down; rotate varieties

**Q:** Can I use nitrogen fertilizer?

**A:** Yes, but excessive nitrogen increases susceptibility

**Q:** Is seed treatment effective?

**A:** Highly effective for seedborne infection control

**Q:** How to manage infected fields?

**A:** Isolate, destroy infected plants, disinfect tools

# POTATO — LATE BLIGHT (PHYTOPHTHORA BLIGHT)

Disease Code: PB001 | Severity: VERY HIGH - Epidemic Potential

## 1. DISEASE OVERVIEW

Late blight is the most economically destructive potato disease worldwide. Caused by water mold (oomycete). Can destroy crops completely within days.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All potato areas, Severe in cool, wet regions

## 2. HISTORY & PAKISTAN CONTEXT

Caused Irish famine (1840s). Entered Pakistan in 1960s. Major epidemics in 1976, 1988, 2005. Responsible for 500 million USD losses globally annually.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                |
|-----------------|----------------------------------------|
| Scientific Name | Phytophthora infestans (Mont.) de Bary |
| Disease Code    | PB001                                  |
| Common Names    | N/A                                    |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne sporangia; rain-splash water; contaminated seed pieces; overwintering in tubers

**Optimal Conditions:** Temperature: 12-18°C (optimal); High humidity (>80%); Persistent leaf wetness (12+ hours); Cool, wet weather

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                             |
|----------------|--------------------------------------------------------------------------------------|
| Early Stage    | Water-soaked spots on older leaves; white fungal growth on leaf undersides           |
| Moderate Stage | Spots rapidly enlarge; affected leaves curl and blacken; white mycelium proliferates |
| Severe Stage   | Complete plant collapse; all foliage destroyed; tubers develop brown rotting         |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties: Setanta, Nicola, Cara, Acoustic (available in Pakistan)
- Eliminate volunteer potato plants
- Spray fungicides preventively before disease appearance
- Recommended fungicides: Mancozeb (800g/ha), Metalaxyl+Mancozeb (2kg/ha), Dimethomorph (800ml/ha)
- Spray every 7-10 days during favorable weather
- Use certified seed from reliable sources

- Avoid overhead irrigation
- Proper field sanitation

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** How fast does late blight spread?

**A:** Exponential spread possible; under optimal conditions, entire field can be infected in 3-5 days

**Q:** Can I control it without fungicides?

**A:** Very difficult; resistance in susceptible varieties is unreliable

**Q:** How long is withholding period?

**A:** Typically 5-7 days; check product label

**Q:** Is tuber infection preventable?

**A:** Prevention difficult once leaves are infected; fungicides slow spread but tubers may still get infected

**Q:** What resistance genes are available?

**A:** R genes (R1-R11); combine multiple R genes for durable resistance

# POTATO — EARLY BLIGHT (ALTERNARIA BLIGHT)

Disease Code: PB002 | Severity: HIGH

## 1. DISEASE OVERVIEW

Early blight affects potato crops throughout growing season. Characterized by target-spot lesions on older leaves.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All potato areas

## 2. HISTORY & PAKISTAN CONTEXT

Present in Pakistan since introduction of potato cultivation. Less economically important than late blight but still significant.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                               |
|-----------------|-------------------------------------------------------|
| Scientific Name | Alternaria solani (Ell. & G. Martin) L. Jones & Grout |
| Disease Code    | PB002                                                 |
| Common Names    | N/A                                                   |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; rain-splash dispersal; contaminated crop debris; seedborne

**Optimal Conditions:** Temperature: 25-27°C (optimal); Warm, humid conditions; Free leaf moisture helps spore germination

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                               |
|----------------|----------------------------------------------------------------------------------------|
| Early Stage    | Small circular brown spots on older leaves; distinctive concentric rings (target-like) |
| Moderate Stage | Spots enlarge; yellowing around lesions; spots with yellow margin and brown rings      |
| Severe Stage   | Extensive defoliation; premature leaf drop; tuber quality affected                     |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Remove lower leaves as disease appears
- Apply fungicides: Chlorothalonil (500ml/ha), Mancozeb (800g/ha)
- Spray at 7-10 day intervals starting 4-6 weeks after planting
- Use resistant varieties where available
- Destroy infected foliage
- Adequate spacing for air circulation

- Proper fertilization - avoid excessive nitrogen

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Is early blight preventable?

**A:** Easily managed with timely fungicide application; less severe than late blight

**Q:** When should I start spraying?

**A:** Begin preventive sprays 4-6 weeks after planting or at first disease appearance

**Q:** Can I use same fungicide as late blight?

**A:** Some products control both; confirm on label

**Q:** Is yield loss significant?

**A:** 10-30% yield loss if uncontrolled; more quality than quantity issue

**Q:** How does weather affect timing?

**A:** After rain, monitor closely and spray within 2 days

# COTTON — COTTON LEAF CURL DISEASE

Disease Code: CLC001 | Severity: VERY HIGH - Economic Threat

## 1. DISEASE OVERVIEW

CLCV caused 20 billion USD loss (500 million bales) from 1992-1997 in Pakistan. Major production constraint. Transmitted by Bemisia whitefly.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All cotton growing areas, Particularly severe in Sindh, Punjab

## 2. HISTORY & PAKISTAN CONTEXT

First detected in Pakistan in 1989. Caused catastrophic epidemic in 1990s. Still major threat despite management efforts. Pakistani strains are extremely virulent.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                               |
|-----------------|-------------------------------------------------------|
| Scientific Name | Cotton Leaf Curl Virus (CLCuD) - Whitefly-transmitted |
| Disease Code    | CLC001                                                |
| Common Names    | N/A                                                   |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Whitefly (Bemisia tabaci) vector; no seed transmission; persistent virus

**Optimal Conditions:** High whitefly populations; warm temperature (25-35°C); irrigation stress; high nitrogen

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                                       |
|----------------|------------------------------------------------------------------------------------------------|
| Early Stage    | Upward leaf curling; vein thickening; chlorotic mottling                                       |
| Moderate Stage | Severe leaf curl; stunting of plant growth; abnormal leaf veining                              |
| Severe Stage   | Leaves remain curled; severe stunting; yield reduction 50-100%; complete crop failure possible |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties: FH-1000, FH-114, IUB-222
- Control whitefly population: Yellow sticky traps, neem oil spray
- Eliminate alternative hosts (weeds, volunteer plants)
- Early planting to escape late-season whitefly build-up
- Avoid excessive nitrogen - encourages whitefly reproduction
- Remove and destroy infected plants
- Avoid planting near other cotton fields

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Can infected plants recover?

**A:** No; virus is persistent; infected plants must be destroyed

**Q:** Is resistant variety immunity permanent?

**A:** Resistance can break down; monitor new virus strains

**Q:** Can I spray insecticides to control whitefly?

**A:** Yes, but whiteflies develop resistance; rotate insecticide classes

**Q:** When is critical planting period?

**A:** Early planting (March-April) before whitefly population build-up

**Q:** What is the role of weeds?

**A:** Weeds are virus reservoir; eliminate all alternate hosts

# COTTON — ANGULAR LEAF SPOT

Disease Code: CLC002 | Severity: MODERATE to HIGH

## 1. DISEASE OVERVIEW

Bacterial leaf spot causing angular lesions on cotton leaves, bolls, and stems.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Humid cotton regions, KP, upper Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Long-standing problem in Pakistan. Becomes severe during monsoon season.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                |
|-----------------|----------------------------------------|
| Scientific Name | Xanthomonas axonopodis pv. malvacearum |
| Disease Code    | CLC002                                 |
| Common Names    | N/A                                    |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Contaminated seed; windblown rain; infected crop debris; insect damage

**Optimal Conditions:** High humidity; Free leaf moisture; Temperature: 25-30°C; Rain damage and insect wounds

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                               |
|----------------|--------------------------------------------------------|
| Early Stage    | Small angular lesions on leaves with yellow halo       |
| Moderate Stage | Lesions expand; cotton bolls develop dark brown spots  |
| Severe Stage   | Extensive leaf spotting; boll rot; premature boll drop |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties: NIAB-999, Sitara-2019
- Hot water seed treatment (50°C for 30 min)
- Apply copper-based bactericides: Copper Oxychloride (2.5 kg/ha)
- Ensure good drainage
- Avoid overhead irrigation
- Destroy infected plant parts

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Is seed treatment alone sufficient?

**A:** No; additional field management is necessary

**Q:** When to apply fungicides?

**A:** Preventively during monsoon season

**Q:** Can weather prediction help?

**A:** Yes; monitor rainfall forecasts and increase vigilance

# TOMATO — LATE BLIGHT

Disease Code: TB001 | Severity: HIGH

## 1. DISEASE OVERVIEW

Late blight affects tomato similar to potato. More common in cool season crops and hill stations.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Cool regions: KP, northern Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Periodically severe in cool regions of Pakistan. Epidemic years: 2008, 2014, 2020.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                |
|-----------------|----------------------------------------|
| Scientific Name | Phytophthora infestans (Mont.) de Bary |
| Disease Code    | TB001                                  |
| Common Names    | N/A                                    |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne sporangia; rain-splash; contaminated seed; infected debris

**Optimal Conditions:** Temperature: 12-18°C; High humidity; Persistent leaf wetness

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                         |
|----------------|------------------------------------------------------------------|
| Early Stage    | Water-soaked spots on leaves with indistinct margins             |
| Moderate Stage | Spots enlarge rapidly; white mycelium on undersides; leaves wilt |
| Severe Stage   | Complete defoliation; stem cankers; fruit rot                    |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Spray preventive fungicides: Mancozeb (800g/ha), Metalaxyl+Mancozeb
- Avoid overhead irrigation
- Remove infected foliage immediately
- Ensure proper spacing
- Use copper-based fungicides for organic management

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** When does late blight affect tomato?

**A:** Cool season (November-March) is high-risk period

**Q:** Can I use same fungicide as late blight potato?

**A:** Yes; many multi-crop fungicides available

**Q:** Is fruit affected?

**A:** Yes; brown sunken lesions on fruits make them unmarketable

# TOMATO — EARLY BLIGHT (ALTERNARIA LEAF SPOT)

Disease Code: TB002 | Severity: HIGH

## 1. DISEASE OVERVIEW

Early blight is the most common tomato leaf disease in Pakistan. Causes target-spot lesions similar to potato.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All tomato growing regions

## 2. HISTORY & PAKISTAN CONTEXT

Present wherever tomatoes are grown in Pakistan. Economically significant throughout growing season.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                               |
|-----------------|-------------------------------------------------------|
| Scientific Name | Alternaria solani (Ell. & G. Martin) L. Jones & Grout |
| Disease Code    | TB002                                                 |
| Common Names    | N/A                                                   |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; rain-splash; contaminated debris; seedborne infection

**Optimal Conditions:** Temperature: 25-27°C (optimal); Warm, humid conditions; 8-12 hour leaf wetness

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                         |
|----------------|------------------------------------------------------------------|
| Early Stage    | Circular lesions with concentric rings on lower leaves           |
| Moderate Stage | Lesions enlarge; leaves yellow and drop; fruit may develop spots |
| Severe Stage   | Premature defoliation; fruit exposed to sunscald; fruit rot      |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Remove lower leaves regularly as preventive measure
- Prune suckers and maintain good canopy structure for air circulation
- Apply fungicides: Chlorothalonil (500ml/ha), Mancozeb (800g/ha)
- Spray every 7-10 days starting from first disease symptoms
- Use drip irrigation to avoid wetting foliage
- Mulching reduces soil splash

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Should I remove all lower leaves?

**A:** Remove leaves 12 inches above soil once disease appears; preventively remove 4-6 leaves weekly

**Q:** How many sprays are needed?

**A:** Average 8-12 sprays throughout growing season depending on weather

**Q:** Can I control without fungicides?

**A:** Good management practices reduce severity but fungicides are usually needed

**Q:** What about fruit quality?

**A:** Defoliation causes sunscald on fruits; prevent through good canopy management

# TOMATO — BACTERIAL SPOT

Disease Code: TB003 | Severity: MODERATE to HIGH

## 1. DISEASE OVERVIEW

Bacterial disease causing small, dark lesions on leaves, stems, and fruits.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All regions, Particularly Sindh, southern Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Present in Pakistan since 1980s. Economically important, especially affecting fruit quality.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                   |
|-----------------|---------------------------|
| Scientific Name | Xanthomonas euvesicatoria |
| Disease Code    | TB003                     |
| Common Names    | N/A                       |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Windblown rain; insect damage; contaminated tools; seedborne

**Optimal Conditions:** Temperature: 25-30°C; High humidity; Free moisture on leaves

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                |
|----------------|-------------------------------------------------------------------------|
| Early Stage    | Small waterlogged lesions on leaves with yellow halo                    |
| Moderate Stage | Lesions develop brown centers; fruit develops spots with raised centers |
| Severe Stage   | Extensive leaf spotting; unmarketable fruit; stem cankers               |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use disease-free seed from reliable sources
- Avoid overhead irrigation
- Apply copper-based bactericides: Copper Oxychloride (2.5 kg/ha)
- Sanitize greenhouse and propagation materials
- Remove infected plants immediately
- Rotate fields; avoid planting near infected areas

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Can I save infected plants?

**A:** No; bacteria are systematic; remove infected plants

**Q:** Is resistance available?

**A:** Some tolerance in certain varieties; complete resistance not available

**Q:** How to prevent via seed?

**A:** Use certified disease-free seed; seed treatment with hot water

# SUGARCANE — RED ROT

Disease Code: SR001 | Severity: VERY HIGH

## 1. DISEASE OVERVIEW

Red rot is the most destructive sugarcane disease in Pakistan and South Asia. Causes internal reddening of cane tissue.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All sugarcane areas, Particularly Punjab and Sindh

## 2. HISTORY & PAKISTAN CONTEXT

Major disease since 1920s in India/Pakistan. Caused severe epidemics. Still major production constraint.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                       |
|-----------------|-------------------------------|
| Scientific Name | Colletotrichum falcatum Went. |
| Disease Code    | SR001                         |
| Common Names    | N/A                           |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Contaminated seed cane; water; insects; weather-related

**Optimal Conditions:** Temperature: 25-30°C; Moderate rainfall; Warm, humid conditions

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                          |
|----------------|-----------------------------------------------------------------------------------|
| Early Stage    | Leafspot on upper leaves with elliptical red margins                              |
| Moderate Stage | Cane internodes show red discoloration when cut; striped pattern                  |
| Severe Stage   | Complete reddening of cane; internal rot; reduced juice quality and sugar content |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties: CP-77-400, Thatta-10, SCSSH-62
- Use disease-free seed cane from certified sources
- Seed treatment: Dip seed pieces in Bordeaux mixture (1%) for 10 minutes
- Spray fungicide: Bordeaux mixture (1%) at 4-5 month stage
- Avoid waterlogging and poor drainage
- Intercrop with legumes to reduce inoculum
- Destroy diseased plants immediately

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** How can I identify red rot at harvest?

**A:** Cut cane cross-section; infected cane shows red discoloration

**Q:** Is seed treatment essential?

**A:** Highly recommended; prevents seedborne transmission

**Q:** Can I use old seed cane?

**A:** No; use certified disease-free seed; stored seed deteriorates

**Q:** How does weather affect severity?

**A:** High rainfall increases infection; manage water carefully

**Q:** What is impact on sugar recovery?

**A:** 3-5% loss in sugar recovery in infected canes

# SUGARCANE — LEAF SCALD

Disease Code: SR002 | Severity: HIGH

## 1. DISEASE OVERVIEW

Bacterial disease causing linear white streaking on sugarcane leaves.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Warm sugarcane areas, Sindh, southern Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Emerged as significant problem in 1970s. Cyclical epidemics in warm years.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                |
|-----------------|----------------------------------------|
| Scientific Name | Xanthomonas albilineans (Ashby) Dowson |
| Disease Code    | SR002                                  |
| Common Names    | N/A                                    |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Contaminated seed cane; insect vectors; infected water

**Optimal Conditions:** High temperature (>28°C); Wounds facilitate infection; Overhead moisture

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                      |
|----------------|---------------------------------------------------------------|
| Early Stage    | Fine yellow or reddish lines on leaf blades                   |
| Moderate Stage | Lines widen; blades develop white-gray streaks; leaves wither |
| Severe Stage   | Premature leaf death; shoot blight; plant stunting            |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use disease-free seed material
- Disinfect harvesting equipment
- Plant resistant varieties: CP-43-72, Thatta-10
- Avoid wounding during cultivation
- Eliminate infected plants

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Is this disease as damaging as red rot?

**A:** Different impact; red rot affects quality, leaf scald affects plant health

**Q:** How to disinfect equipment?

**A:** 1% Bleach solution or alcohol wipes between fields

**Q:** Can antibiotics control it?

**A:** Antibiotics not practical in field; prevention is key

# MAIZE — COMMON RUST

Disease Code: MR001 | Severity: MODERATE to HIGH

## 1. DISEASE OVERVIEW

Rust appears as reddish-brown pustules on corn leaves, mainly on upper surfaces.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All maize areas, Particularly Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Common problem in Pakistan maize production. Occasionally causes significant yield loss.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                 |
|-----------------|-------------------------|
| Scientific Name | Puccinia sorghi Schwein |
| Disease Code    | MR001                   |
| Common Names    | N/A                     |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne uredospores; wind-dispersed; alternate hosts involved

**Optimal Conditions:** Temperature: 15-22°C; High humidity; Free moisture on leaves

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                  |
|----------------|---------------------------------------------------------------------------|
| Early Stage    | Small circular lesions on leaves                                          |
| Moderate Stage | Reddish-brown pustules arranged in rows; may appear on both leaf surfaces |
| Severe Stage   | Heavy pustule development; leaf tissue dies; premature senescence         |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant hybrids available from seed companies
- Apply fungicides if needed: Propiconazole (25% EC) at 800ml/ha
- Spray at VT to R1 growth stage if >10% leaf area infected
- Proper nitrogen management
- Field sanitation

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** How much yield loss possible?

**A:** Generally 5-15% in susceptible hybrids; resistant hybrids largely avoid losses

**Q:** Can I use same fungicide as wheat rust?

**A:** Yes; many products work on both crop rusts

**Q:** When to start monitoring?

**A:** From tasseling stage onwards at weekly intervals

# MAIZE — NORTHERN CORN LEAF BLIGHT (NCLB)

Disease Code: MR002 | Severity: MODERATE

## 1. DISEASE OVERVIEW

NCLB causes elongated lesions on corn leaves, primarily affecting middle and lower leaves.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Cool maize areas, KP and northern Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Occasional problem in cooler maize-growing regions of Pakistan.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                      |
|-----------------|----------------------------------------------|
| Scientific Name | Exserohilum turcicum (Pass.) Leonard & Suggs |
| Disease Code    | MR002                                        |
| Common Names    | N/A                                          |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; rain-splash; overwinter on crop residue

**Optimal Conditions:** Cool, wet conditions; Temperature: 16-24°C; High humidity; Leaf wetness

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                 |
|----------------|--------------------------------------------------------------------------|
| Early Stage    | Watersoaked lesions at leaf tip areas                                    |
| Moderate Stage | Lesions elongate along leaf veins; gray-brown appearance; halo yellowing |
| Severe Stage   | Extensive leaf blight; premature defoliation                             |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties
- Apply fungicides preventively in cool, wet seasons
- Destroy crop residue after harvest
- Avoid overhead irrigation
- Adequate spacing

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Is this mainly in northern areas?

**A:** Yes; prefers cool temperatures; rare in hot plains

**Q:** Do I need to spray if no symptoms?

**A:** Prevention sprays recommended in susceptible areas during cool, wet periods

**Q:** How does this differ from southern corn leaf blight?

**A:** Different pathogen; NCLB is more prevalent in Pakistan

# BELL PEPPER — BACTERIAL SPOT

Disease Code: BP001 | Severity: MODERATE

## 1. DISEASE OVERVIEW

Small, dark, raised lesions on pepper leaves and fruits.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Warm regions, Sindh, southern Punjab

## 2. HISTORY & PAKISTAN CONTEXT

Emerging problem in Pakistan pepper production.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                   |
|-----------------|---------------------------|
| Scientific Name | Xanthomonas euvesicatoria |
| Disease Code    | BP001                     |
| Common Names    | N/A                       |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Contaminated seed; windblown rain; insect damage

**Optimal Conditions:** Warm, humid conditions; Free leaf moisture; Temperature: 25-30°C

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                         |
|----------------|------------------------------------------------------------------|
| Early Stage    | Small watersoaked lesions with yellow halo on leaves             |
| Moderate Stage | Lesions develop brown centers; fruit develops raised brown spots |
| Severe Stage   | Extensive spotting makes fruit unmarketable                      |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use disease-free seed
- Apply copper-based bactericides
- Avoid overhead irrigation
- Remove infected plants
- Rotate fields

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** How can I prevent this disease?

**A:** Use disease-free seed and avoid overhead irrigation

**Q:** Is it curable once established?

**A:** No; focus on prevention and containment

**Q:** Can resistant peppers help?

**A:** Some tolerance in certain varieties available

# MANGO — ANTHRACNOSE (LEAF BLIGHT)

Disease Code: MG001 | Severity: HIGH

## 1. DISEASE OVERVIEW

Most important mango disease in Pakistan. Causes black spots on leaves, panicles, and fruits.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All mango areas

## 2. HISTORY & PAKISTAN CONTEXT

Present wherever mangoes are grown. Causes pre-harvest and post-harvest losses.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                        |
|-----------------|--------------------------------|
| Scientific Name | Colletotrichum gloeosporioides |
| Disease Code    | MG001                          |
| Common Names    | N/A                            |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; rain-splash; contaminated tools; insect damage

**Optimal Conditions:** High humidity; Temperature: 20-28°C; Free leaf and fruit moisture

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                                        |
|----------------|---------------------------------------------------------------------------------|
| Early Stage    | Small dark lesions on young leaves                                              |
| Moderate Stage | Lesions enlarge; cause necrosis; flower panicles blacken; fruit spotting begins |
| Severe Stage   | Extensive fruit rot; unmarketable fruit; reduced fruit set                      |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Spray carbendazim (500g/ha) or chlorothalonil (500ml/ha) at flower bud appearance
- Repeat spray every 10-14 days during flowering and fruiting
- Remove infected flowers and fruits
- Improve drainage; avoid waterlogging
- Prune to improve canopy ventilation
- Post-harvest fungicide dip for export fruit

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** When is the critical spray period?

**A:** Flowering to early fruit stage is most critical

**Q:** Can infected fruits be sold locally?

**A:** Yes if disease is superficial; deep rot makes them unsuitable

**Q:** How many sprays are needed?

**A:** Typically 3-4 sprays during flowering-fruiting stage

# MANGO — POWDERY MILDEW

Disease Code: MG002 | Severity: MODERATE

## 1. DISEASE OVERVIEW

White powdery growth on mango leaves, flowers, and young fruits.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Northern mango orchards

## 2. HISTORY & PAKISTAN CONTEXT

Common in northern mango-growing regions during spring.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details           |
|-----------------|-------------------|
| Scientific Name | Oidium mangiferae |
| Disease Code    | MG002             |
| Common Names    | N/A               |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; wind-dispersed

**Optimal Conditions:** Temperature: 20-26°C; Low to moderate humidity; Dry conditions favor disease

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                  |
|----------------|-----------------------------------------------------------|
| Early Stage    | White powder on leaf undersides                           |
| Moderate Stage | Powder spreads to both surfaces; flower clusters affected |
| Severe Stage   | Extensive powdery coating; premature fruit drop           |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Spray sulfur (80% WP) at 2kg/ha
- Apply fungicides: Propiconazole or Triazole class
- Avoid excessive nitrogen
- Improve air circulation
- Remove infected parts

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Why sulfur is preferred for mango?

**A:** Safe for fruit, less damage, effective in low humidity

**Q:** Can I use same products as on crops?

**A:** Yes; but test on small area first due to phytotoxicity concerns

**Q:** When to stop spraying?

**A:** After fruit set stage; post-harvest dips if needed for export

# APPLE — POWDERY MILDEW

Disease Code: AP001 | Severity: MODERATE to HIGH

## 1. DISEASE OVERVIEW

White powdery coating on apple leaves, flowers, and fruits.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Northern apple regions, Himachal, Gilgit

## 2. HISTORY & PAKISTAN CONTEXT

Major problem in northern apple orchards of Pakistan.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                          |
|-----------------|--------------------------------------------------|
| Scientific Name | Podosphaera leucotricha (Ell. & Ev.) E.S. Salmon |
| Disease Code    | AP001                                            |
| Common Names    | N/A                                              |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne conidia; overwintering in buds; wind-dispersed

**Optimal Conditions:** Temperature: 15-20°C; Dry conditions; Moderate humidity

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                 |
|----------------|----------------------------------------------------------|
| Early Stage    | White powder on young leaves and shoots                  |
| Moderate Stage | Extensive powdery growth; shoots curl; flowers affected  |
| Severe Stage   | Fruit cracking; russet appearance; reduced fruit quality |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Spray sulfur (80% WP) at 2-3kg/ha
- Apply fungicides: Propiconazole (25% EC) at 800ml/ha
- Spray before disease appearance
- Prune infected shoots
- Avoid excessive nitrogen fertilization

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** When to start sulfur sprays?

**A:** Early spring before bud break; continue every 10-14 days

**Q:** Can sulfur damage apple fruit?

**A:** No; safe for apple; monitor temperature (avoid  $>27^{\circ}\text{C}$  application)

**Q:** Is pruning necessary?

**A:** Yes; remove severely infected shoots before spraying

# APPLE — APPLE SCAB

Disease Code: AP002 | Severity: HIGH

## 1. DISEASE OVERVIEW

Fungal disease causing olive-brown scabs on apple leaves and fruits.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Northern high-rainfall areas

## 2. HISTORY & PAKISTAN CONTEXT

Major problem in wet years in northern Pakistan.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                               |
|-----------------|---------------------------------------|
| Scientific Name | Venturia inaequalis (Cooke) G. Winter |
| Disease Code    | AP002                                 |
| Common Names    | N/A                                   |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Ascospores from infected leaf litter; rain-splash; airborne conidia

**Optimal Conditions:** Cool, wet conditions; Temperature: 7-24°C; High humidity; Free moisture on leaves

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                  |
|----------------|-----------------------------------------------------------|
| Early Stage    | Velvety olive-brown spots on leaves                       |
| Moderate Stage | Spots spread; cracks in fruit skin; russetting of fruit   |
| Severe Stage   | Distorted fruit; significant fruit drop; leaf defoliation |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Destroy infected fallen leaves
- Spray lime sulfur or fungicides preventively
- Apply mancozeb (800g/ha) or myclobutanil (400ml/ha)
- Improve drainage
- Pruning for better air circulation

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** Why is leaf litter management critical?

**A:** Ascospores overwinter in fallen leaves; elimination breaks disease cycle

**Q:** When are critical spray periods?

**A:** Bud break through fruit set stage; high risk during wet springs

**Q:** Can scabby fruit be used?

**A:** Not for export; cosmetic damage reduces market value significantly

# CHICKPEA — ASCOCHYTA BLIGHT

Disease Code: CK001 | Severity: HIGH

## 1. DISEASE OVERVIEW

Fungal disease causing brown lesions with concentric rings on chickpea leaves and pods.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: All chickpea areas, Particularly Punjab and Sindh

## 2. HISTORY & PAKISTAN CONTEXT

Major chickpea disease in Pakistan. Causes epidemics in wet years.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                        |
|-----------------|--------------------------------|
| Scientific Name | Ascochyta rabiei (Pass.) Labr. |
| Disease Code    | CK001                          |
| Common Names    | N/A                            |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Seed-borne; rain-splash; airborne conidia

**Optimal Conditions:** Cool, wet weather; Temperature: 15-22°C; High humidity; Free moisture

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                                            |
|----------------|---------------------------------------------------------------------|
| Early Stage    | Small gray lesions on leaves with dark borders and concentric rings |
| Moderate Stage | Lesions enlarge; affect petioles and stems; plants wilt             |
| Severe Stage   | Extensive defoliation; pod rot; seed discoloration                  |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use disease-free seed
- Seed treatment: Thiram or Carbendazim (3g/kg seed)
- Crop rotation: 2-3 year minimum with non-legume crops
- Avoid overhead irrigation
- Destroy infected crop residue
- Apply fungicide if needed: Mancozeb (800g/ha)

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** How important is seed treatment?

**A:** Critical; seed-borne inoculum is major infection source

**Q:** Can I grow chickpea year after year?

**A:** No; 2-3 year rotation minimum to break disease cycle

**Q:** Is fungicide effective?

**A:** Yes; timely application can reduce losses but prevention is better

# LENTIL — RUST (UROMYCES RUST)

Disease Code: LN001 | Severity: MODERATE

## 1. DISEASE OVERVIEW

Rust appears as yellow-orange pustules on lentil leaves.

**Global Significance:** This disease causes significant economic losses in all major crop-producing countries.

**Pakistan Distribution:** Found in: Northern lentil areas

## 2. HISTORY & PAKISTAN CONTEXT

Occasional problem in cooler lentil-growing regions.

## 3. SCIENTIFIC CLASSIFICATION

| Parameter       | Details                                    |
|-----------------|--------------------------------------------|
| Scientific Name | Uromyces lentis (Kulkarni & Hegde) Laundon |
| Disease Code    | LN001                                      |
| Common Names    | N/A                                        |

## 4. DISEASE CYCLE & TRANSMISSION

**Transmission:** Airborne uredospores; wind-dispersed

**Optimal Conditions:** Cool, humid conditions; Temperature: 12-18°C

## 5. DISEASE SYMPTOMS & PROGRESSION

| Severity Stage | Symptoms                                               |
|----------------|--------------------------------------------------------|
| Early Stage    | Yellow-green lesions on leaf undersides                |
| Moderate Stage | Brown rust pustules develop; leaves yellow and senesce |
| Severe Stage   | Extensive defoliation; premature plant death           |

## 6. MANAGEMENT & CONTROL STRATEGIES

- Use resistant varieties where available
- Spray fungicides if severe: Propiconazole (25% EC) at 800ml/ha
- Avoid overhead irrigation
- Proper spacing for air circulation

## 7. FREQUENTLY ASKED QUESTIONS

**Q:** When does rust appear on lentil?

**A:** Cool season lentils are most affected; appears late in season

**Q:** Is seed treatment needed?

**A:** Not critical for rust; important for other seed-borne diseases

**Q:** Can I mix fungicides?

**A:** Yes; rotate different fungicide classes to prevent resistance

## APPENDIX A: GLOSSARY OF TECHNICAL TERMS

**Ascomycota:** A phylum of fungi with spore-bearing structures (asci). Examples: Alternaria, Pyricularia.

**Basidiomycota:** A phylum of fungi including rusts and smuts. Spores develop from basidia.

**Canker:** A localized dead area on stems or branches surrounded by callus tissue.

**Conidium:** An asexual fungal spore produced by conidiophores on hyphae.

**Defoliation:** Premature shedding or dropping of leaves from plants.

**Epidemiology:** Study of disease occurrence and spread patterns in plant populations.

**Fungicide:** Chemical substance that kills or inhibits fungal growth and reproduction.

**Hyphae:** Thread-like filaments that comprise fungal mycelium.

**Inoculum:** Pathogenic material (spores, cells, tissue) initiating infection.

**Lesion:** Localized area of diseased tissue showing discoloration or necrosis.

**Necrosis:** Death of plant tissue appearing as brown or black areas.

**Oomycota:** Water molds; fungal-like organisms including Phytophthora species.

**Pathogen:** Organism causing disease in another organism.

**Pustule:** Fungal fruiting structure from which spores are released (especially rust fungi).

**Resistant variety:** Plant variety with genetic resistance to specific pathogen or disease.

**Saprophyte:** Organism living on dead/decaying organic matter.

**Sporulation:** Production and release of spores by fungus or pathogen.

**Systemic:** Affecting entire plant or spreading internally through vascular system.

**Systemic Fungicide:** Fungicide penetrating plant tissue and distributed throughout plant.

**Withholding Period:** Minimum time between pesticide application and harvest for safety.

# APPENDIX B: SEASONAL DISEASE MANAGEMENT CALENDAR

| Season                  | Months  | High-Risk Diseases                    | Management Actions                                           |
|-------------------------|---------|---------------------------------------|--------------------------------------------------------------|
| Winter<br>(Cool Season) | Nov-Feb | Wheat Rusts, Late Blight, Apple Scab  | Monitor wheat and potato fields; prevent whitefly; manage a  |
| Spring<br>(Transition)  | Mar-May | Blast, Anthracnose, Common Rust       | Monitor rice nurseries; spray mango flowers; check corn rus  |
| Monsoon<br>(Wet Season) | Jun-Sep | Late Blight, Bacterial Spot, Blast    | Increase fungicide frequency; avoid overhead irrigation; ren |
| Autumn<br>(Harvest)     | Oct-Nov | Red Rot, Leaf Scald, Cotton Leaf Curl | Check sugarcane at harvest; scout cotton for whitefly; prep  |

# APPENDIX C: RECOMMENDED FUNGICIDES & BACTERICIDES (Pakistan-Approved)

| Active Ingredient    | Product Name         | Dosage            | Applicable Crops            | Disease Target                   |
|----------------------|----------------------|-------------------|-----------------------------|----------------------------------|
| Mancozeb             | Various brands       | 800g/ha           | Potato, Tomato, Rice, Wheat | Blights, Leaf Spots, Rust        |
| Copper Oxychloride   | Various brands       | 2.5 kg/ha         | All vegetable & fruit crops | Bacterial diseases, Fungal spots |
| Carbendazim          | Bavistin, others     | 500g/ha           | All crops                   | Broad-spectrum fungal control    |
| Propiconazole        | Tilt, Rimi           | 25% EC @ 800ml/ha | Wheat, Corn, Fruit crops    | Rusts, Powdery Mildew            |
| Metalaxyl + Mancozeb | Ridomil Gold         | 2 kg/ha           | Potato, Tomato              | Late Blight (preventive)         |
| Chlorothalonil       | Various brands       | 500ml/ha          | Potato, Tomato, Vegetables  | Multi-disease control            |
| Validamycin A        | Various brands       | 2-3 L/ha          | Rice                        | Blast (curative)                 |
| Triazole Class       | Hexaconazole, others | 500ml/ha          | Multiple crops              | Various fungal diseases          |

**Important:** Always follow label instructions. Rotate fungicide classes to prevent resistance development. Follow withholding periods (typically 5-14 days). Use PPE when applying pesticides.

## APPENDIX D: FIELD MONITORING CHECKLIST

Use this checklist for regular field monitoring (weekly during growing season):

- Visual inspection of leaves (upper and lower surfaces) for lesions, spots, pustules
- Check stem and petioles for cankers or other abnormalities
- Examine fruits/tubers for symptoms of disease
- Record percentage of plants affected and severity (1-5 scale)
- Note weather conditions (rainfall, temperature, humidity) in past 7 days
- Identify any new disease symptoms not previously observed
- Presence of insect damage (wounds facilitate bacterial/fungal infection)
- Irrigation method and frequency (overhead irrigation increases risk)
- Nitrogen fertilizer application timing and amount (affects disease susceptibility)
- Presence of alternate hosts or volunteer plants (disease reservoirs)
- Effectiveness of previous fungicide applications
- Weed control status (weeds can harbor pathogens)